BJT Current Mirror Pre-Lab

This is from Marcel's team. But, as we said to you, we don't currently have a group for the class and couldn't submit it.

Circuit: top of $R_1 = 6.8\,k\Omega$ and R_x at ground (0 V), both emitters at -10 V. Q_1 is diode-connected, Q_2 copies its current.

Assumptions: $V_{\text{BE}} \approx 0.7$ V, $V_{\text{CE,sat}} = 0.2$ V, matched transistors.

Task 1: Bounds on R_x

Reference current

Loop through R_1 and Q_1:

$$i_{\text{ref}} = \frac{0 - V_{\text{BE}} - (-10)}{6.8\,k\Omega} = \frac{9.3}{6800} \approx 1.37\ \text{mA}$$

Copied current

Accounting for base currents:

$$i_{\text{copy}} = i_{\text{ref}} \frac{\beta}{\beta + 2} \approx i_{\text{ref}} \quad (\beta \gg 1)$$

Active-region condition for Q_2

$V_C = -i_{\text{copy}} R_x$ and $V_E = -10$ V, so

$$V_{\text{CE2}} = 10 - i_{\text{copy}} R_x \geq V_{\text{CE,sat}} = 0.2 \quad \Rightarrow \quad R_x \leq \frac{9.8}{i_{\text{copy}}}$$

Result

$$\boxed{0 \leq R_x \lesssim \tfrac{9.8}{1.37\ \text{mA}} \approx 7.2\,k\Omega}$$

- Lower bound: $R_x = 0$ gives $V_{\text{CE2}} = 10$ V, still active (well below V_{CEO}).
- With $\beta = 100$: $i_{\text{copy}} \approx 1.34$ mA, so $R_{x,\,\text{max}} \approx 7.3\,k\Omega$.
- Above $R_{x,\,\text{max}}$, Q_2 saturates and $i_{\text{copy}} \approx 9.8/R_x$.

Task 2: Multisim Validation

a) i_{ref}

Hand calculation: **1.37 mA**. Multisim reads **1.38 mA**, which implies $V_{\text{BE}} \approx 10 - 1.38\ \text{m} \times 6.8\ \text{k} \approx 0.62$ V for the 2N3394 model. Good agreement.

> **Note: why $i_{\text{copy}} = 1.52$ mA with $R_x = 1\,k\Omega$**
> $V_{\text{CE1}} \approx 0.62$ V but $V_{\text{CE2}} \approx 8.5$ V. The Early effect makes Q_2 conduct about 10% more. This is what step b) removes.

b) Remove the Early effect

For **both** Q1 and Q2 (keep them matched):

1. Double-click transistor > **Value** tab > **Edit model**.
2. Set `VAF = 1e6` (Early voltage) and `BF = 1e6` (β).

3. Click **Change part model**.

Expected: $i_{\text{copy}} \approx i_{\text{ref}} \approx 1.38$ mA.

Result ($R_x = 1\,k\Omega$, emitter supply -10 V): $i_{\text{ref}} = 1.38$ mA, $i_{\text{copy}} = 1.359$ mA. The remaining 1.5% gap is base-current error (β finite).

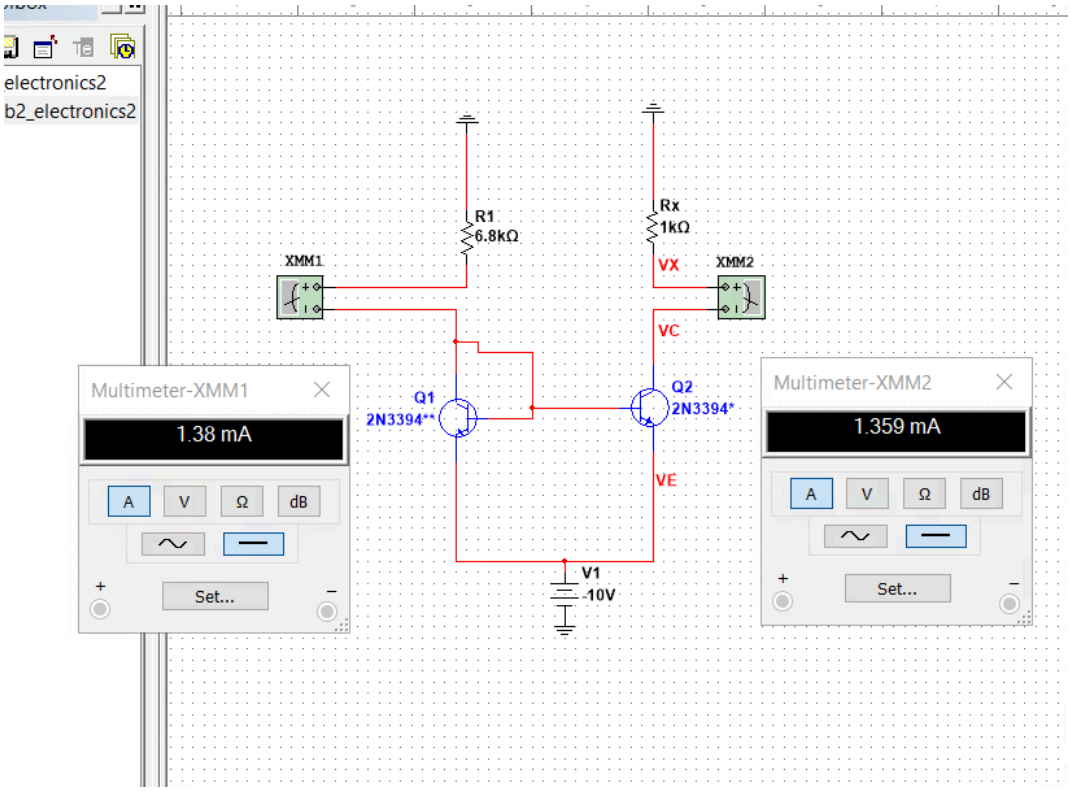


Figure 1: Multisim current mirror circuit

c) Parameter sweep of R_x

1. **Simulate > Analyses and simulation > Parameter Sweep**.
2. Device type: **Resistor**, Name: **Rx**, Parameter: **resistance**.
3. Sweep: linear, 0 to 10 kΩ (e.g. 100 points).
4. Analysis to sweep: **DC Operating Point**.
5. Output: `I(Rx)` (or V_C).

Expected: i_{copy} flat at about 1.38 mA up to about 7.1 kΩ, then dropping as $\approx 9.8/R_x$ (saturation). The knee should match Task 1.

Result: i_{copy} stays constant at about 1.36 mA up to about 7 kΩ, then its magnitude falls as Q_2 saturates. This matches the Task 1 bound ($R_x \lesssim 7.2\,k\Omega$). The negative sign is only the reference direction of `I(RX)`.

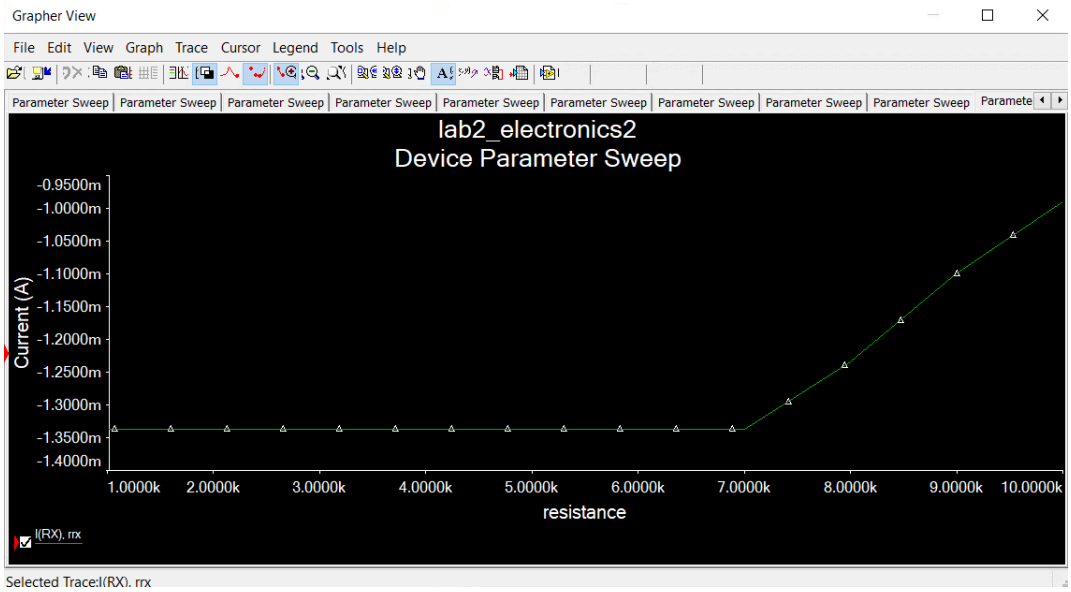


Figure 2: Parameter sweep of R_x

LTspice Version

Recreation of the Multisim circuit: R1 = 6.8k, Rx = 1k, Q1 diode-connected, V1 puts both emitters at −10 V. The 2N3394 model is copied from Multisim (VAF = 2e6), included inline so the file runs on its own.

Netlist

Save as `current_mirror.cir`, then **File** > **Open** in LTspice and press **Run**.

```
* BJT current mirror recreation of Multisim circuit for fun ;)
R1 0 B 6.8k
Rx 0 VC {Rx}
Q1 B B VE QN
Q2 VC B VE QN
V1 VE 0 -10
.param Rx=1k
.model QN NPN(IS=6.101e-14 BF=121.3 NF=1 VAF=2e6 IKF=0.09779 ISE=1.222e-13 NE=1.305
BR=0.2359 NR=1)
.op
.step param Rx 1k 10k 100
* Log list: Icopy and VCE2 at chosen Rx values
.meas OP Iref FIND I(R1) AT=1k
.meas OP Icopy_1k FIND I(Rx) AT=1k
.meas OP Icopy_4k FIND I(Rx) AT=4k
.meas OP Icopy_7k FIND I(Rx) AT=7k
.meas OP Icopy_7k2 FIND I(Rx) AT=7.2k
.meas OP Icopy_8k FIND I(Rx) AT=8k
.meas OP Icopy_10k FIND I(Rx) AT=10k
.meas OP VCE2_7k2 FIND V(VC)-V(VE) AT=7.2k
.meas OP VCE2_8k FIND V(VC)-V(VE) AT=8k
* Knee: Rx where Q2 reaches VCE,sat = 0.2 V
.meas OP Rx_knee WHEN V(VC)-V(VE)=0.2
.end
```

Node map: `B` = Q1 collector + both bases, `VC` = Q2 collector, `VE` = both emitters.

```
Files loaded:
C:\Desktop\MyNotes\uOttawa\Year_4\current_mirror.cir

iref: I(R1) =0.00137996987905 at 1000
icopy_1k: I(Rx) =0.00133690459188 at 1000
icopy_4k: I(Rx) =0.00133690191433 at 4000
icopy_7k: I(Rx) =0.00133689923678 at 7000
icopy_7k2: I(Rx) =0.00133689248469 at 7200
icopy_8k: I(Rx) =0.00123453885317 at 8000
icopy_10k: I(Rx) =0.000991007196717 at 10000
vce2_7k2: V(VC)-V(VE) =0.374374389648 at 7200
vce2_8k: V(VC)-V(VE) =0.123688697815 at 8000
rx_knee: V(VC)-V(VE)=0.2 AT 7374.32908044
```

- **Log:** open **View** > **SPICE Error Log** (Ctrl+L). Scroll past the "Direct Newton" lines to the `Iref`, `Icopy` and `VCE2` tables, one row per Rx step. The knee is where `VCE2` reaches about 0.2 V.
- To see the Early effect (2a), set `VAF=74` and rerun: the flat region now slopes.
- To remove base-current error, set `BF=1e6`: the flat level rises to about 1.38 mA.